

Which experiments deserve a larger compute budget?

OFDG, KXRCF gating, and a deliberately small numerical study

This brief supports experiment selection with a supervisor. It is an initial measured preview, not a completed reproduction or a definitive ranking of methods. Every plot comes from the accompanying isolated preview dataset.

Experiment	Configuration	Question
Smooth advection	P1-P3; N=32-256; T=1.1	Accuracy across order
Piecewise advection	P2; N=128/256; T=1.1	Oscillations versus smearing
Burgers	P2; smooth T=0.6, shock T=2.2	Nonlinear transition to a shock
Lax	P1/192, P2/128, P3/96; T=1.3	Resolution at equal unknown count
Shu-Osher	P2; N=200/400; T=1.8	Shock-wave interaction
Straight/curved transport	P2; 8 ² /16 ² /32 ² ; T=1	Geometry and smooth accuracy
2D Riemann	P2; 32 ² /64 ² ; T=0.25	Wave structure and positivity

Comparison protocol

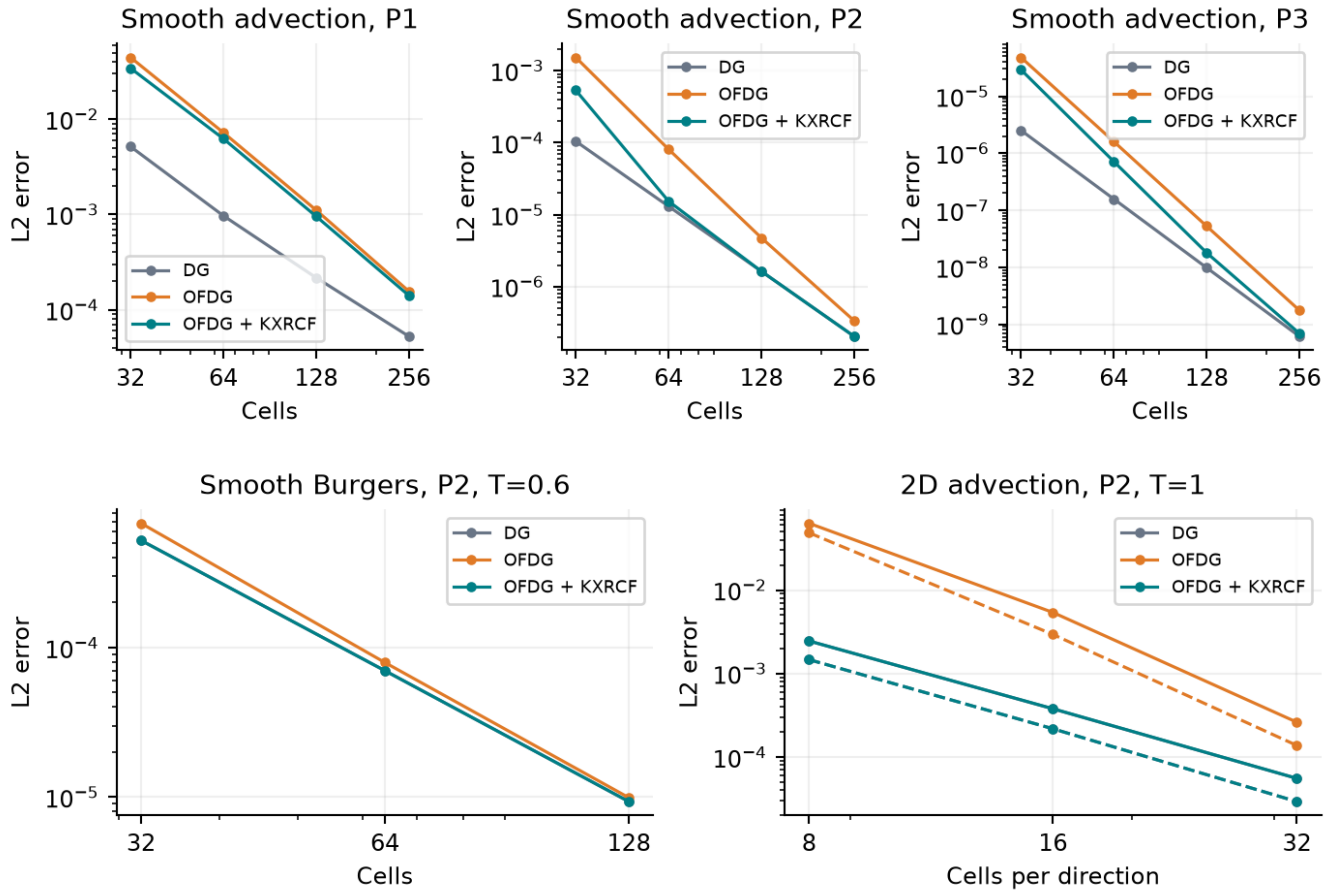
DG, OFDG, and OFDG + KXRCF appear throughout. OEDG is included only on the finer affine scalar-shock and Shu-Osher cases and P2 Lax. All use RK4 and the same CFL within each problem (0.15 scalar, 0.30 Euler). OFDG uses post-step filtering; OEDG uses stage filtering. Euler runs share the positivity limiter. This compares these configured methods, not filtering cadence in isolation.

Measured resource use

103 of 103 solver runs completed successfully; 0 failed or were skipped. Solver and reference wall time: **7.3 minutes**. Sampled peak process-tree RSS: **118 MiB**. Runs were sequential, with one rank in 1D and up to two in 2D on an eight-processor laptop. No timing repetitions were performed.

Limits: 4 GB aggregate resident memory per check; five minutes per check; no new jobs after 45 minutes; 60-minute total ceiling. Memory is sampled and can miss short-lived peaks. Build, analysis, and document preparation are outside reported solver time.

Smooth accuracy and curved geometry



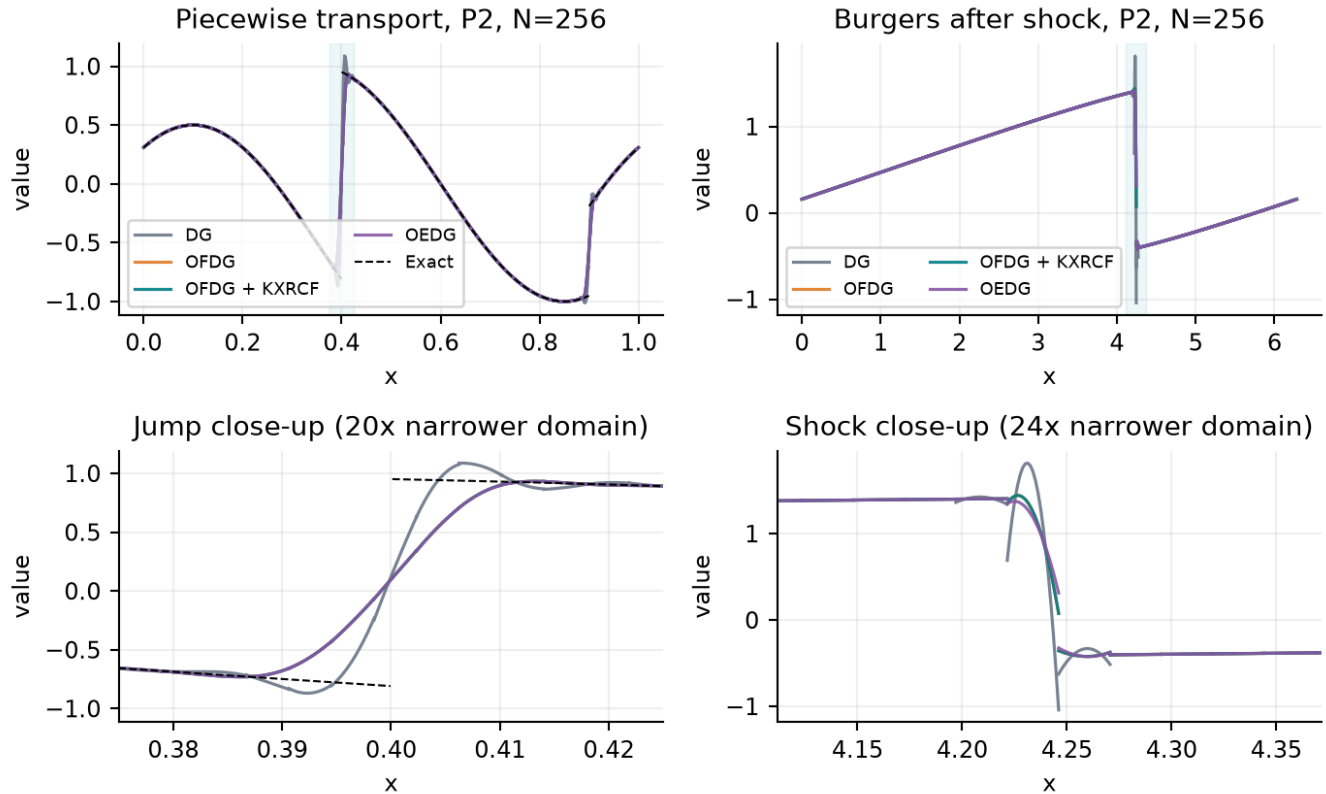
Last-pair L2 slope	Advection P1	P2	P3	Curved P2
DG	2.05	3.00	4.00	2.77
OFDG	2.83	3.83	4.90	4.37
OFDG + KXRCF	2.77	3.00	4.70	2.77

Solid curves in the lower-right plot use the cubic curved mesh; dashed curves use the straight mesh. The unit-square transport speed is (0.7, 0.3), with $\sin^2(\pi(x+y))$ data. This is a paper-inspired adaptation, not the exact 2021 setup. Slopes on these grids are observations, not proof of an asymptotic rate.

Finest-P3 CFL halving changed the reported L2 error by: DG 0.0006%, OFDG 0.0029%, OFDG + KXRCF 0.0003%. A change above 5% triggers a smaller-CFL P3 series. This check compares error norms, not the full solution difference.

At the finest P2 grids, gating nearly matches DG's smooth-error level; ungated OFDG has larger errors. The largest reported periodic scalar conservation drift is $4e-14$.

Scalar shocks: retain the full polynomial



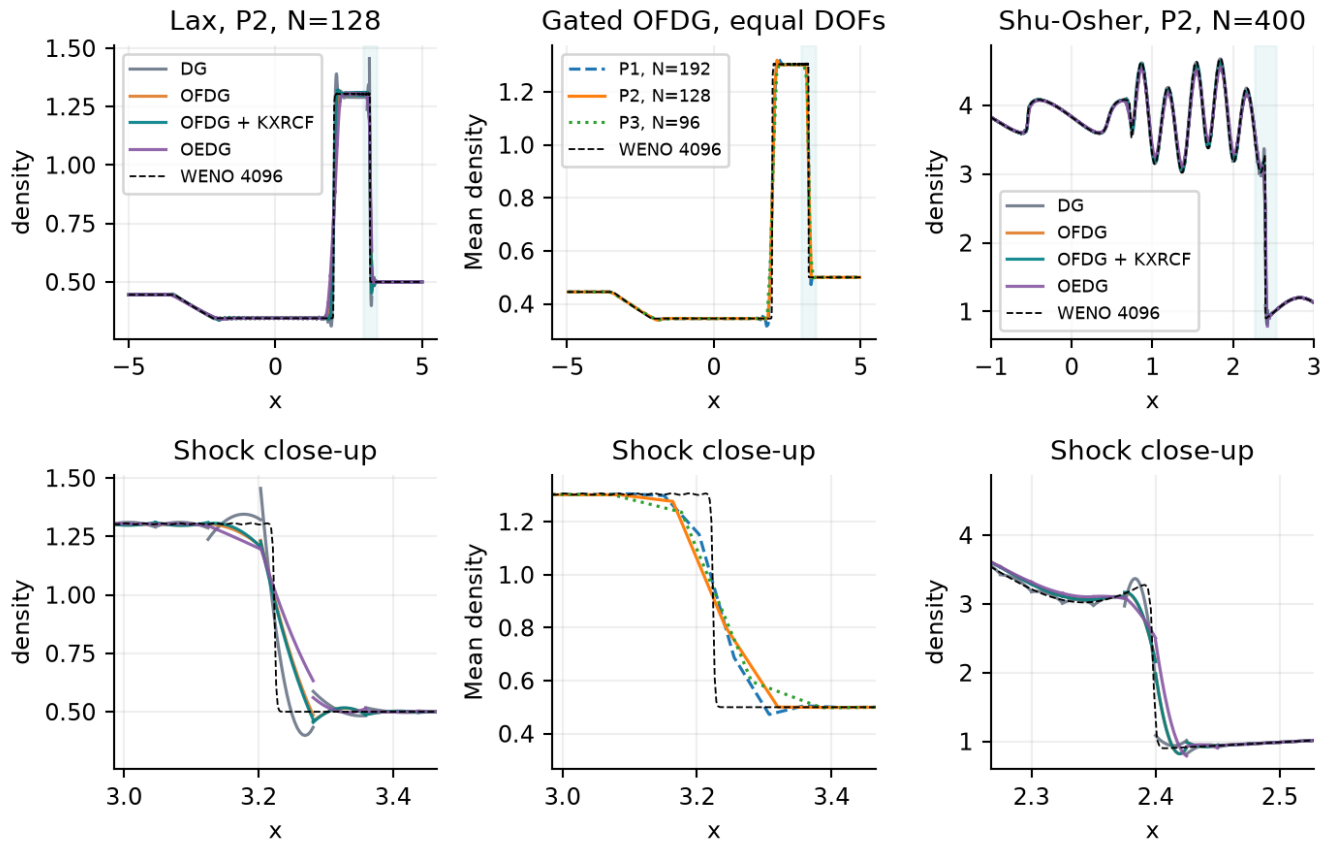
Lines show 21 samples per DG cell, broken at cell interfaces. Cell-center values can hide polynomial oscillations. The piecewise transport has an exact translated reference; post-shock Burgers is shown qualitatively without an independent entropy-solution error estimate.

Case / method	Minimum	Maximum	Undershoot	Overshoot
Transport / DG	-1.01	1.09	0.0119	0.136
Transport / OFDG	-1	0.932	4.18e-08	0
Transport / OFDG + KXRCF	-1	0.933	5.77e-08	0
Transport / OEDG	-1	0.932	0	0
Burgers / DG	-1.04	1.82	0.541	0.317
Burgers / OFDG	-0.422	1.44	0	0
Burgers / OFDG + KXRCF	-0.422	1.44	0	0
Burgers / OEDG	-0.424	1.4	0	0

Extrema are sampled from the full polynomial, not certified continuous extrema. Overshoot and undershoot use the analytic initial-data range: transport $[-1, \sin(0.6\pi)]$ and Burgers $[-0.5, 1.5]$. Sampling and discontinuous initial projection can affect these measurements.

Shaded overview bands mark the close-up intervals. OFDG and gated OFDG nearly overlap on these scalar shocks; the close-ups expose differences that the full-domain plots hide.

Euler: shocks and small-scale structure



Shaded overview bands identify the shock close-ups below them. The Lax order comparison uses 384 scalar unknowns (1,152 conservative unknowns) in every case. Equal unknown count does not imply equal work. The middle panel shows density derived from actual conservative cell averages; the outer panels show the solution polynomials.

Independent reference sensitivity

Reference pair	Mean density difference	Maximum difference
lax 2048 vs 4096	0.00101	0.12
shu osher 2048 vs 4096	0.00324	0.463

References use the separate finite-volume WENO5 / SSPRK3 solver. The 4,096-cell conservative densities are averaged onto the nested 2,048-cell grid for this check. Nonzero differences, especially near shocks, demonstrate that the reference is not exact. No reference-based convergence rate is claimed.

Positivity and interpretation

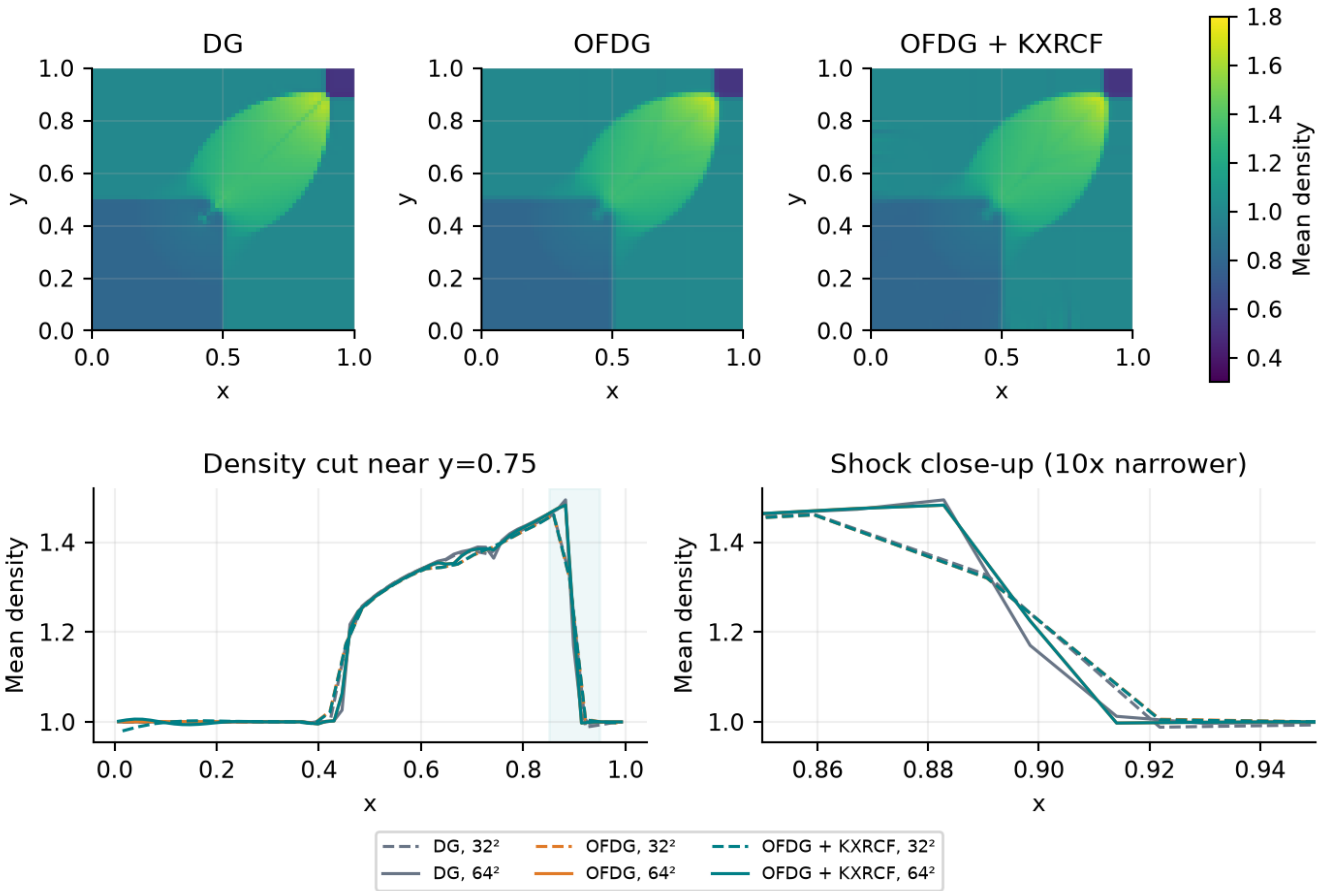
Among successful 1D Euler runs, the smallest reported final density is 0.252 and pressure is 0.361. Positivity checks and limiter counts are recorded in the raw results. A successful exit is not evidence of accurate shock resolution.

At P2/N=400, Shu-Osher triggered 6,252 cumulative positivity cell-limit events for DG, 3 for each OFDG variant, and 2 for OEDG. This is a robustness observation under the common limiter, not a cost or accuracy ranking.

Lax and Shu-Osher retain fixed far-field exterior states. Their domain totals can change through boundary fluxes; the logged total change is not interpreted as a conservation error. The positivity limiter and any rejected steps are part of the comparison, not hidden repairs.

A small 2D Riemann preview

P2 Euler at $T=0.25$. Density maps use 64×64 cells and a shared color scale. Four-quadrant data and outflow boundaries follow the first Riemann setup in the 2024 OEDG paper. These grids are intentionally coarse.



Method / cells	Min. density	Min. pressure	Limited cells*	Rejected steps
DG / 32 ²	0.38	0.207	122	0
OFDG / 32 ²	0.513	0.379	0	0
OFDG + KXRCF / 32 ²	0.513	0.379	0	0
DG / 64 ²	0.431	0.236	482	0
OFDG / 64 ²	0.515	0.384	0	0
OFDG + KXRCF / 64 ²	0.515	0.384	0	0

*Limited cells is the cumulative count reported by the limiter, not the number of distinct cells. The row comparison samples the cell-center row nearest $y=0.75$; its physical position differs between grids. The close-up uses these same cell averages, not polynomial traces. Maps show unsmoothed means; a shared color scale may clip extremes.

What to discuss before scaling up

Suggested next experiments

Priority	Larger experiment	Decision it would support
1	Refine Shu-Osher with reference sensitivity and repeated timings	Does gating retain useful waves at comparable cost?
2	Refine straight and curved smooth transport, then curved Euler	Separate geometry effects from shock-filter effects
3	Refine this 2D Riemann problem with matched boundary conditions	Do differences persist as wave structures resolve?
Later	Blast waves, 3D, and parallel scaling	Only after selecting the scientific question and obtaining suitable hardware

Limits of this preview

The selected configurations, fluxes, initialization, time integrator, filtering cadence, and positivity treatment all influence the outcome. No universal method ranking follows from this matrix. Only smooth P3 advection receives a time-step sensitivity check. Shock profiles need broader spatial refinement and error measures. Native NURBS geometry is outside this preview; curved results use supported polynomial geometry.

Failures and omissions: All scheduled solver runs completed; no cases were dropped.

Reproducibility

The preview directory contains the manifest, exact commands, source fingerprint and patch, MFEM/project revisions, checkpointed results, raw logs, rank-local samples, reference CSVs, convergence table, analysis summary, and figures. The runner resumes only when its source and executable fingerprint matches. Use a new output directory after code changes. Existing full-study and thesis assets are unchanged.

Project revision: f50a6dff8f98; MFEM revision: a1ce49fb5742. Source/executable SHA-256: 494e40a9d0d4532cd9cb7226...

Research basis

[1] J. Lu, Y. Liu, C.-W. Shu (2021), An Oscillation-Free Discontinuous Galerkin Method for Scalar Hyperbolic Conservation Laws. Scalar advection Examples 1(a,b) and Burgers Example 2.

[2] Y. Liu, J. Lu, C.-W. Shu (2022), An Essentially Oscillation-Free Discontinuous Galerkin Method for Hyperbolic Systems. Lax and Shu-Osher examples, equal-DOF comparisons and full-polynomial plots.

[3] M. Peng, Z. Sun, K. Wu (2024), OEDG: Oscillation-Eliminating Discontinuous Galerkin Method for Hyperbolic Conservation Laws. First four-quadrant Riemann configuration. Local copies are in the repository papers folder.